Estimation and Dynamics in Household Diseases

Casper Woroszylo
MBI, The Ohio State University

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Introduction

Estimation

Dynamics

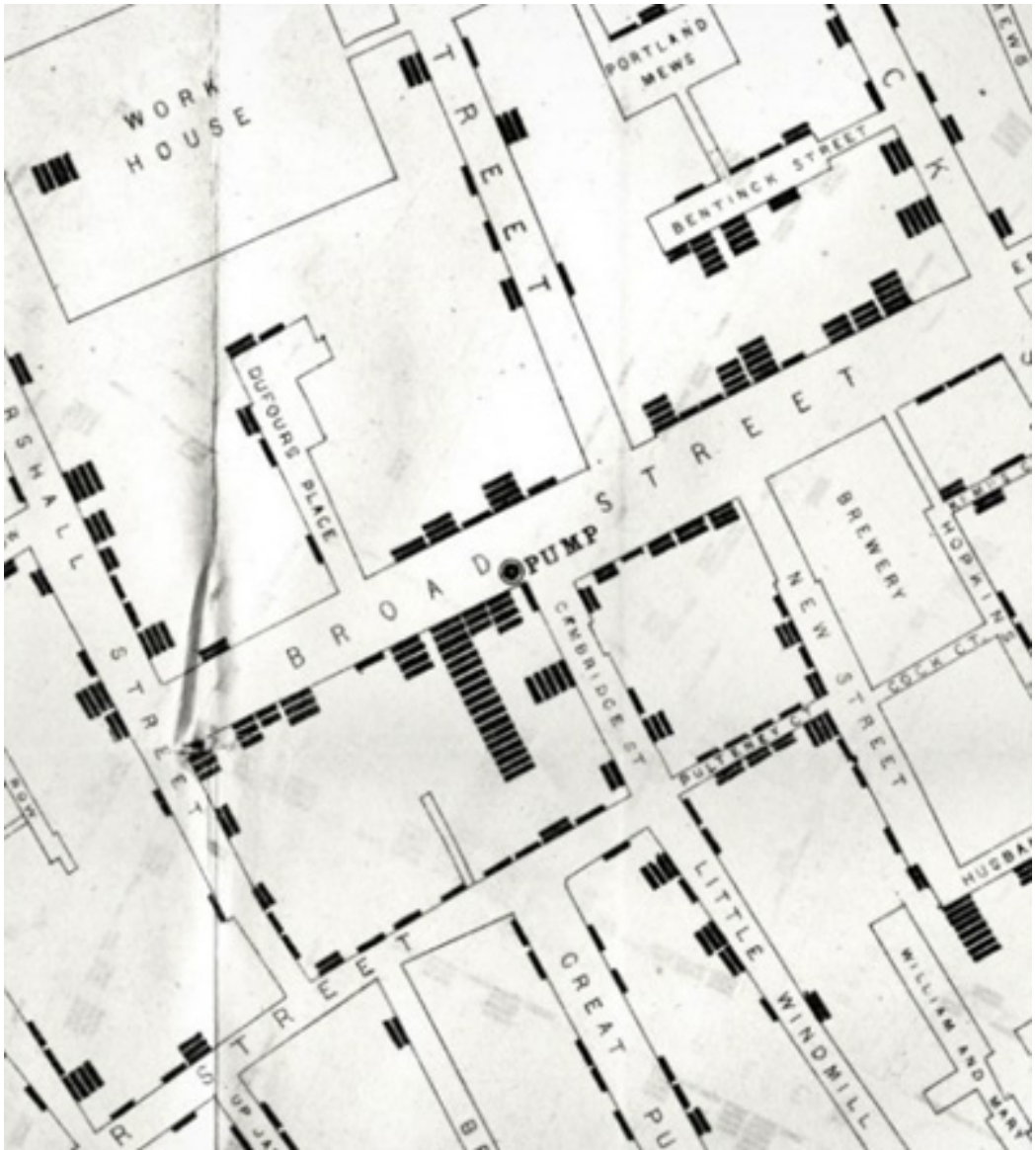
MCMC

Thanks

Introduction

Cholera outbreak of 1854

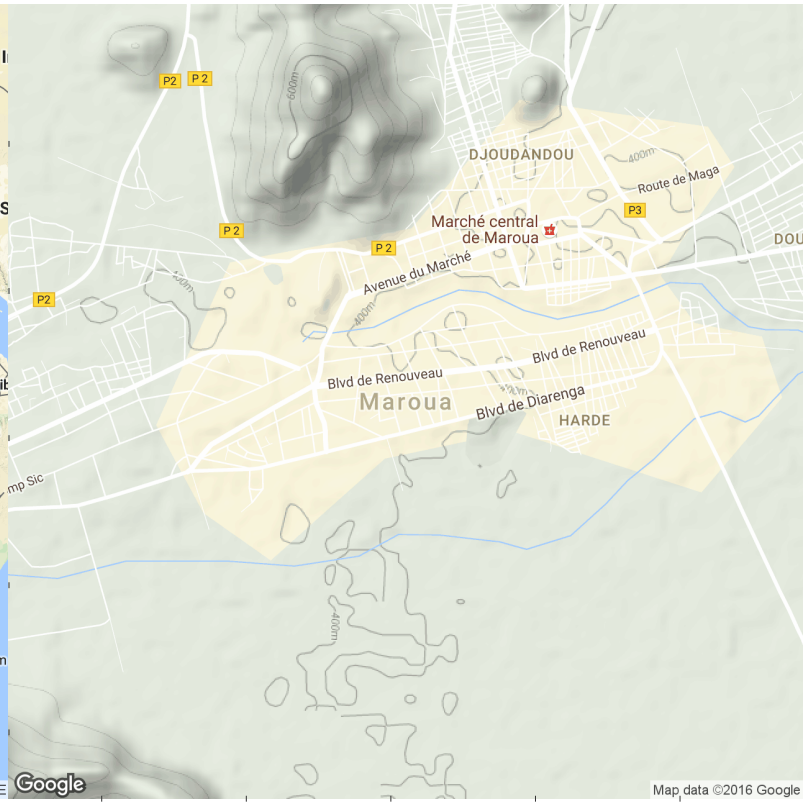
- Originally thought Cholera was transmitted via air (miasma)
- John Snow showed that Cholera was a water-born disease



Motivation

- 1.9 million deaths from diarrhoeal illness in 2011
- Majority of deaths occurred in children under the age of five in developing countries
- Kills more children annually than malaria
- Second leading cause of mortality among children

Marou, Cameroon



Data overview

Marou, Cameroon [1]

- Surveyed 120 households (785 individuals)
- Collected June 1 to August 1 (2013)
- Asked if experienced gastrointestinal symptoms during the 30 days prior to the survey

How does microbiome diversity affect susceptibility to pathogens (wrt environment)?

[1] Profitos, Tien, et al., Int. J. Environ. Res. Public Health, 2014

Setup

Cross-sectional data of the following nature:

	I	J	A	DI	DJ	DA	V
household 1:	0	0	2	0	0	0	0
household 2:	0	1	2	0	0	1	0
household 3:	0	1	7	0	0	0	0
household 4:	0	4	6	0	0	0	1
household 5:	1	3	5	0	1	3	0

At the household level we have

1. Number of healthy Infants (I), Juveniles (J), Adults (A)
2. Number of diseased Infants (D_I), Juveniles (D_J), Adults (D_A)
3. Pathogen status (V)

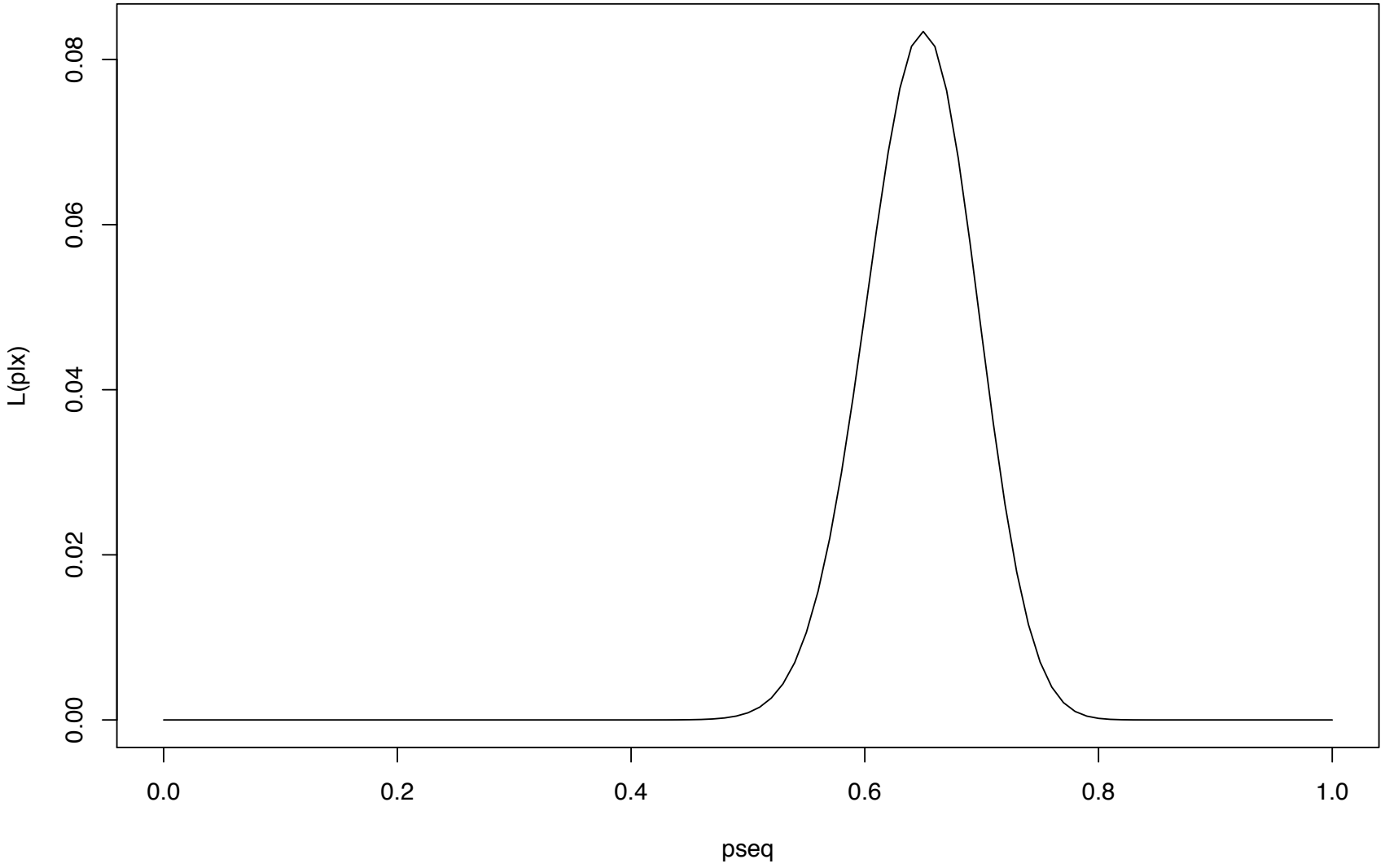
Probability of disease in a given compartment?

Estimation

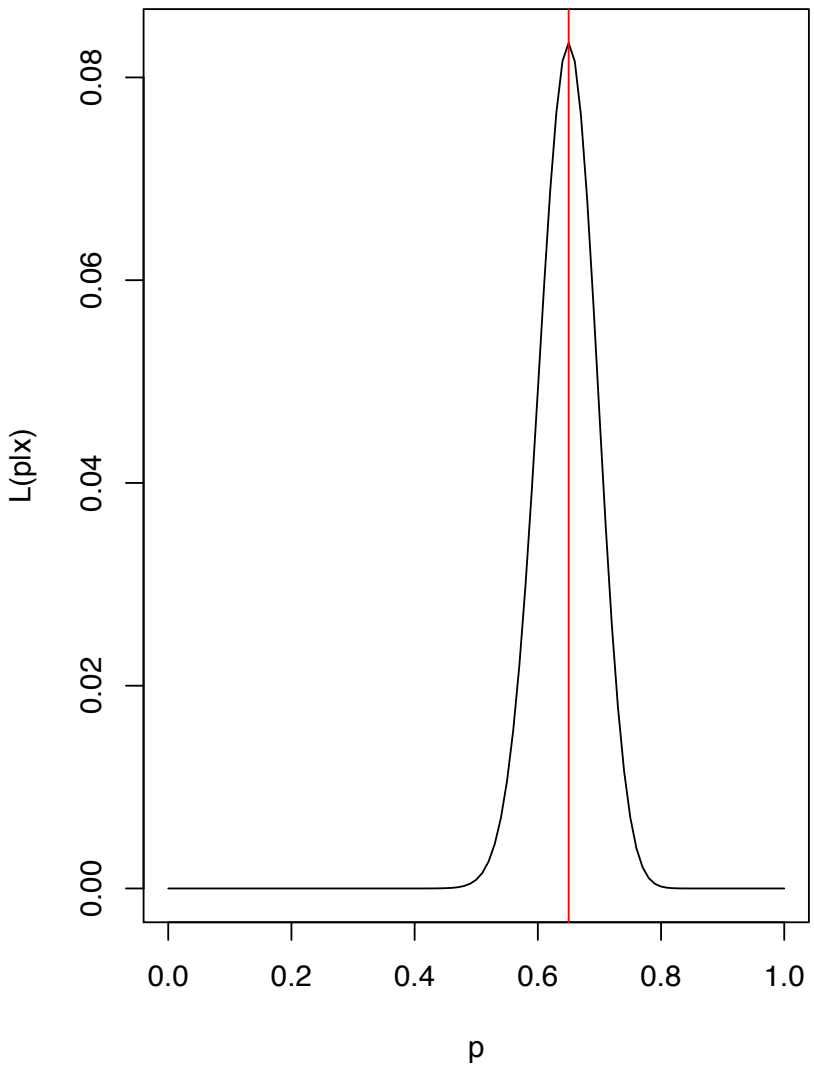
Likelihood

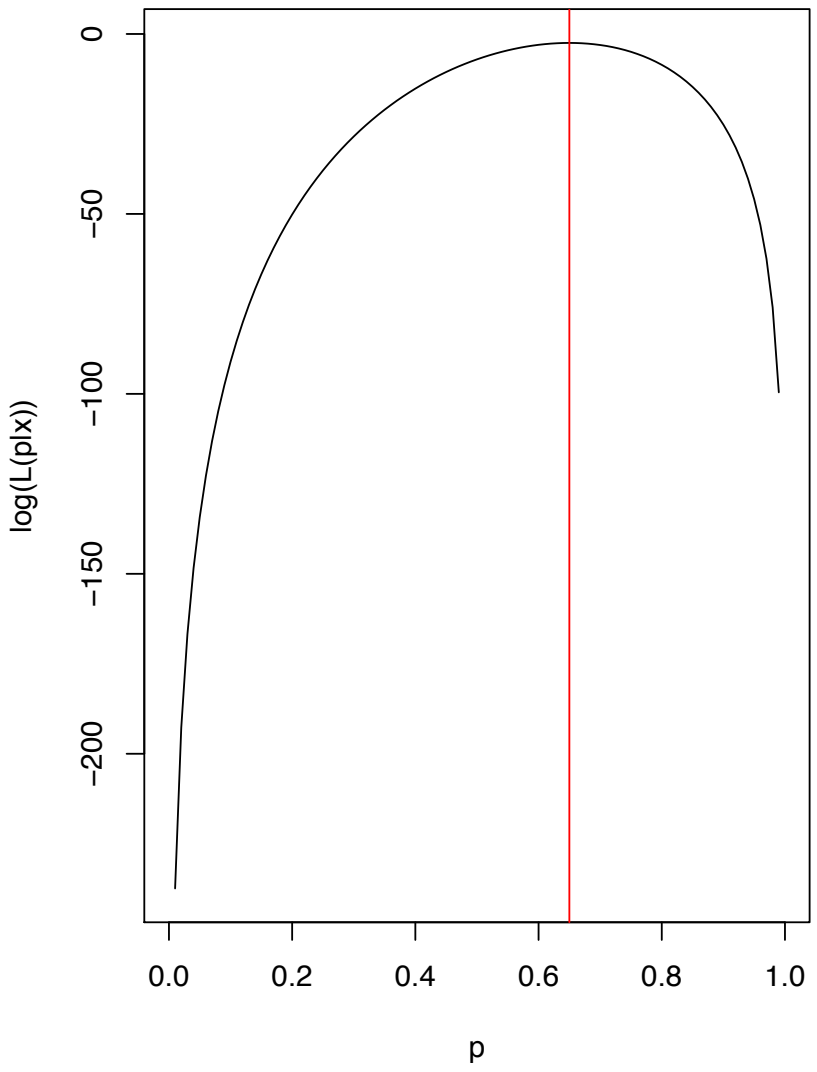
e.g. Observe 65 heads in 100 tosses of coin (probability of heads p):

$$L(p|x) = \binom{100}{65} p^{65}(1-p)^{35}$$



Maximum Likelihood





Maximum Likelihood Estimate

$$\log(L(p|x)) = \log\binom{20}{5} + 65\log(p) + 35\log(1-p)$$

$$\frac{\partial}{\partial p}\log(L(\theta|x)) = \frac{65}{p} - \frac{35}{1-p}$$

Hence, $\hat{p}_{\text{MLE}} = 65/100$

Household structure

64 households with water contamination status:
For $k \in \{I, J, K\}$:

$$L(\theta|x) = \prod_{k \in \{I,J,A\}} P[D_k = x_k | N_k = x_k + x_{Dk}, V = x_V] \times$$
$$P[N_k = x_k + x_{Dk}] P[V = x_V]$$

Assume $D_k|(N_k, V) \sim \text{Binomial}(N_k, p_k(V))$ and $N_k \sim \text{Poisson}(\lambda_k)$.
We use an empirical mean for V.

Likelihood Estimation

Assume you observe M households. Calculations for $p_k(V), \lambda_k, k \in \{I, J, A\}, V = 0, 1$, are identical. For simplicity, we express the estimates for $p(V), \lambda$:

$$
\begin{aligned}
L(p(V), \lambda) &= \prod_{k=1}^{M} \binom{n_k}{x_k} p(V)^{x_k} (1 - p(V))^{n_k - x_k} \frac{\lambda^{n_k} \exp(-\lambda)}{n_k!} \\
\ell(p(V), \lambda) &= \log(L(p(V), \lambda)) \\
&= -\sum_{k=1}^{M} \log\left(\frac{1}{x_k!(n_k - x_k)!}\right) + \log(p(V)) \sum_{k=1}^{M} x_k \\
&\quad + \log(1 - p(V)) \sum_{k=1}^{M} (n_k - x_k) \\
&\quad + \log(\lambda) \sum_{k=1}^{M} n_k - M\lambda
\end{aligned}
$$

Maximum Likelihood Estimates

The derivatives of the log-likelihood are

$$\frac{\partial}{\partial \lambda}\ell(p(V),\lambda) = \frac{\sum_{k=1}^{M} n_k}{\lambda} - M$$
$$\frac{\partial}{\partial p(V)}\ell(p(V),\lambda) = \frac{\sum_{k=1}^{M} x_k}{p(V)} - \frac{\sum_{k=1}^{M}(n_k - x_k)}{1-p(V)}$$

$$\hat{p}(V) = \frac{\sum_{\{k:v_k=V\}} x_k}{\sum_{\{k:v_k=V\}} n_k} = \hat{E}[X_k|V]$$
$$\hat{\lambda} = \frac{\sum_{k=1}^{M} n_k}{M}$$

Estimates from data

	I	J	A
p(1)	0.00	0.174	0.067
p(0)	0.00	0.094	0.110
lambda	0.27	2.254	4.873

Is age structure / environment relevant?

Likelihood ratio

Two hypothesis tests

1. Age structure: $H_0 : p_k(V) = p_k(V)$ vs $H_1 : p_k(V) = p(V)$
2. Environment: $H_0 : p_k(V) = p_k(V)$ vs $H_1 : p_k(V) = p_k$

Likelihood ratio: $\lambda = L_0/L_1$

Creating Households

For $k \in \{I, J, K\}$

1. Generate $V \sim \text{Bernoulli}(p(V))$
2. Sample $N_k \sim \text{Poisson}(\lambda_k)$
3. Sample $D_k \sim \text{Binomial}(N_k, p_k(V))$
4. Repeat to obtain multiple households

Test for Age Structure and Environment

Parametric bootstrapped LRT tests:

1. Generate M households under H_0
2. Using step 1,evaluate λ_i where L_1, L_0 are the likelihoods under the alternate and null hypotheses, respectively.
3. Perform steps 1 and 2 10000 times and and look at the 2.5% and 97.5% percentiles. If the 1 (0 if use log) is in the interval, we do not have sufficient evidence to reject H_0. Otherwise, we reject H_0 and take H_1 as our assumption.

Results

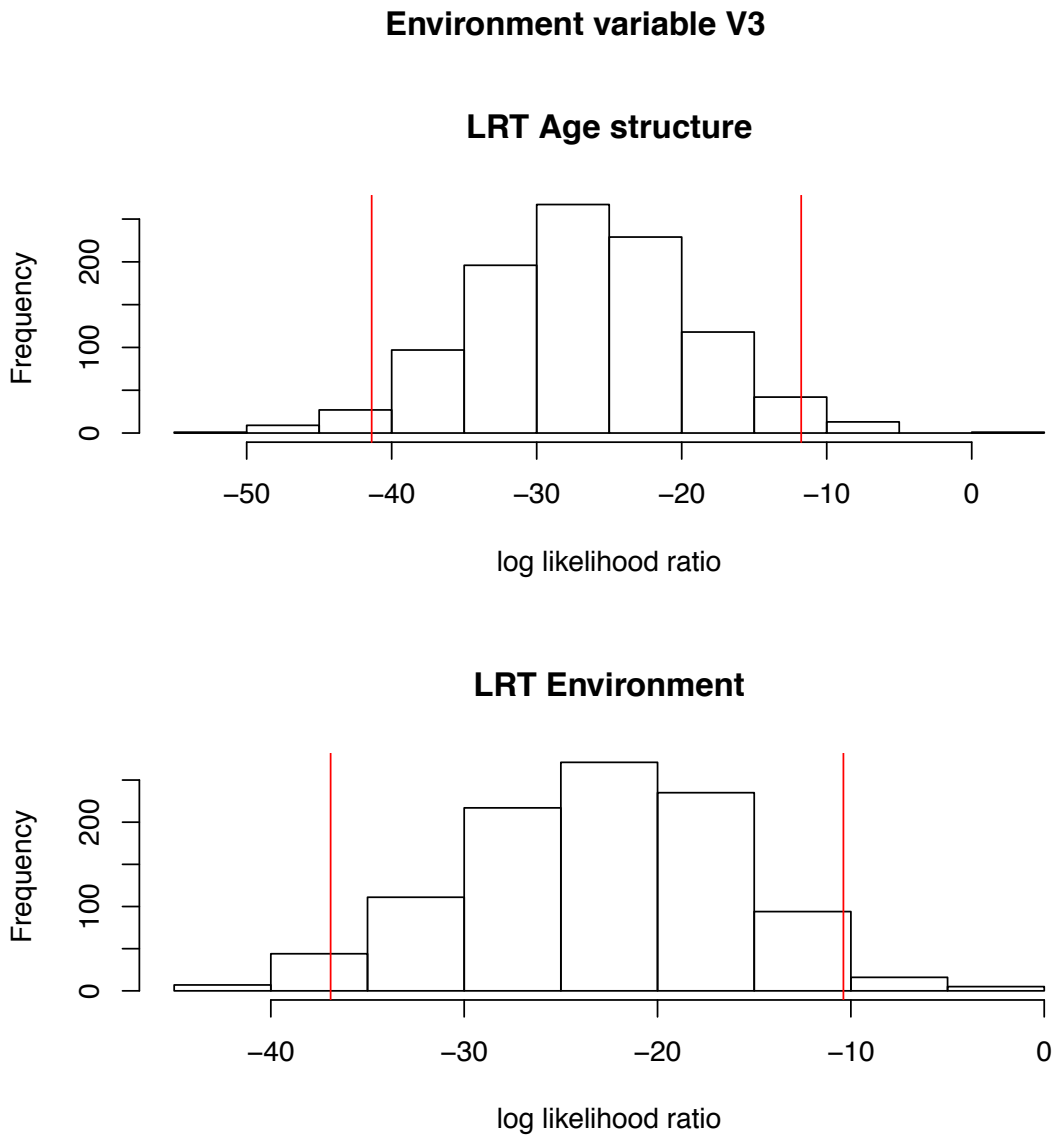


Figure 2

Dynamics

Biochemical Representation

For $(S_i, I_i, D_i, V), i \in \{J, A\}$, we consider the following system:

$$
\begin{aligned}
S_i + I_i &\xrightarrow{\beta_{ii}} 2I_i \\
S_i + I_j &\xrightarrow{\beta_{ij}} I_i + I_j \\
S_i &\xrightarrow{\phi_i V} I_i \\
S_i &\xrightarrow{\alpha_i} D_i \\
I_i &\xrightarrow{\gamma_i - \nu_i} S_i \\
I_i &\xrightarrow{\nu_i} D_i \\
D_i &\xrightarrow{\delta_i} S_i
\end{aligned}
$$

$$\frac{d}{dt}S_1 = -\beta_{12}S_1I_2 - \beta_{11}S_1I_1 - \phi_1S_1V - \alpha_1S_1 + \delta_1D_1 + (\gamma_1 - \nu_1)I_1$$

$$\frac{d}{dt}S_2 = -\beta_{21}S_2I_1 - \beta_{22}S_2I_2 - \phi_2S_2V - \alpha_2S_2 + \delta_2D_2 + (\gamma_2 - \nu_2)I_2$$

$$\frac{d}{dt}I_1 = \beta_{21}S_2I_1 + \beta_{11}S_1I_1 + \phi_1S_1V - \gamma_1I_1$$

$$\frac{d}{dt}I_2 = \beta_{12}S_1I_2 + \beta_{22}S_2I_2 + \phi_2S_2V - \gamma_2I_2$$

$$\frac{d}{dt}D_1 = \alpha_1S_1 + \nu_1I_1 - \delta_1D_1$$

$$\frac{d}{dt}D_2 = \alpha_2S_2 + \nu_2I_2 - \delta_2D_2$$

Problem

What we know:

- Observe $(S_i + I_i), D_i$
- System of ODE's for S_i, I_i, D_i for each environment status V
- Data observed in ~ 2 month period with no major outbreak

What we want:

- S_i, I_i
- Idea of how transmission rates vary

Assumptions

System is in steady state:

$$\frac{d}{dt}S_1 = -\beta_{12}S_1I_2 - \beta_{11}S_1I_1 - \phi_1S_1V - \alpha_1S_1 + \delta_1D_1 + (\gamma_1 - \nu_1)I_1 = 0$$
$$\frac{d}{dt}S_2 = -\beta_{21}S_2I_1 - \beta_{22}S_2I_2 - \phi_2S_2V - \alpha_2S_2 + \delta_2D_2 + (\gamma_2 - \nu_2)I_2 = 0$$
$$\frac{d}{dt}I_1 = \beta_{21}S_2I_1 + \beta_{11}S_1I_1 + \phi_1S_1V - \gamma_1I_1 = 0$$
$$\frac{d}{dt}I_2 = \beta_{12}S_1I_2 + \beta_{22}S_2I_2 + \phi_2S_2V - \gamma_2I_2 = 0$$
$$\frac{d}{dt}D_1 = \alpha_1S_1 + \nu_1I_1 - \delta_1D_1 = 0$$
$$\frac{d}{dt}D_2 = \alpha_2S_2 + \nu_2I_2 - \delta_2D_2 = 0$$

MCMC

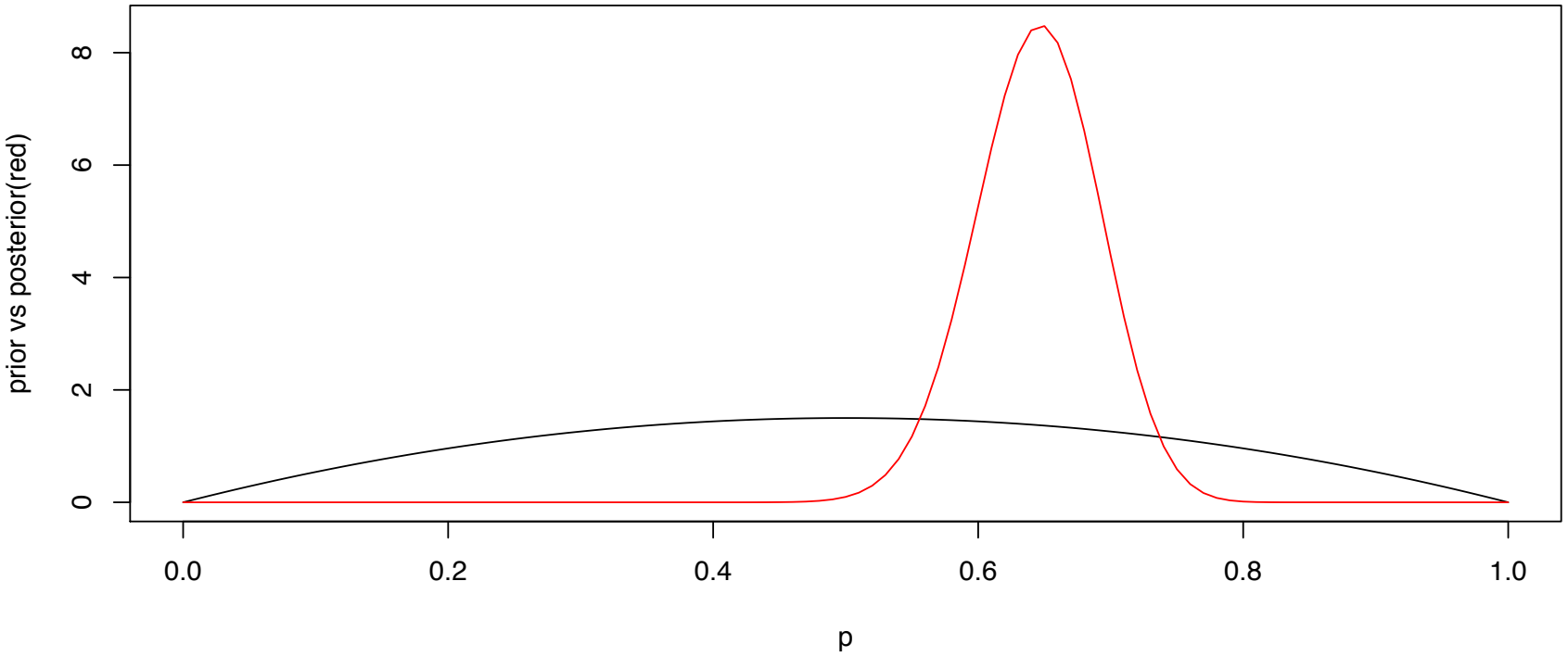
Bayes Theorem

Suppose we put a prior on the probability $p \sim \text{Beta}(2, 2)$ for the coin example and consider the likelihood for the 65 heads out of 100 trials we observed.

$$
\begin{aligned}
p|(2,2) &\sim \text{Beta}(2,2) \\
x|(100,p) &\sim \text{Binomial}(100,p) \\
p|(100,65,2,2) &\propto p^{2-1}(1-p)^{2-1}p^{65}(1-p)^{35} \\
&\propto p^{(2+65)-1}(1-p)^{(2+35)-1}
\end{aligned}
$$

so that $p|(100, 65, 2, 2) \sim \text{Beta}(2 + 65, 2 + 35)$.

Bayes Theorem



Gibbs Sampler

If we have a joint posterior we want to sample from, we can directly sample from the posterior distributions sequentially as a Markov chain until the chain reaches the stationary distribution.

1. Draw $\theta_1^{(t+1)}$ from $p(\theta_1|\theta_2^{(t)})$
2. Draw $\theta_2^{(t+1)}$ from $p(\theta_2|\theta_1^{(t+1)})$
3. Repeat until chain is stationary

Metropolis-Hastings

If the posterior doesn't have a closed form, we use Metropolis Hastings, i.e. we make a guess for a proposal distribution (how the chain does a random walk) and use a likelihood criterion to accept or reject the proposed point.

1. Propose a move to y with probability $q(y|x)$
2. Calculate the ratio

$$r = \frac{p(y)q(x|y)}{p(x)q(y|x)}$$

3. Accept the move with probability

$$\alpha = \min\{1, r\}$$

   or remain at x $(X_{t+1} = X_t)$
4. Repeat until chain is stationary

Metropolis-Hastings - Intuition

$$\pi(y) = \sum_x \pi(x)p(x,y)$$
$$\pi(y)p(y,x) = \pi(x)p(x,y)$$

Then for $x \neq y$

$$\alpha(y|x) = \frac{p(y)q(x|y)}{p(x)q(y|x)}$$
$$p(y|x) = \begin{cases} q(y|x), & \alpha(x,y) \geq 1 \\ q(y|x)\alpha(y|x), & \alpha(x,y) < 1 \end{cases}$$

so that $p(y)p(x|y) = p(x)p(y|x)$.

Functional relationship

Suppose $S_1 + I_1 + D_1 = N_1, S_2 + I_2 + D_2 = N_2$. Let $\tilde{D}_1 = S_1 + I_1$ and $\tilde{D}_2 = S_2 + I_2$. Then, from the ODEs, the theoretical means are

$$
\begin{aligned}
\tilde{D}_1 &= S_1 + I_1 = \left(\frac{\gamma_1}{\beta_{11} I_1 + \beta_{12} I_2 + \phi_1} + 1 \right) I_1 = f_1^{\boldsymbol{\theta}_1}(I_1, I_2) \\
\tilde{D}_2 &= S_2 + I_2 = \left(\frac{\gamma_2}{\beta_{22} I_2 + \beta_{21} I_1 + \phi_2} + 1 \right) I_2 = f_2^{\boldsymbol{\theta}_2}(I_1, I_2) \\
\tilde{D}_1 &= \frac{(\alpha_1 - \nu_1) I_1 + \delta_1 D_1}{\alpha_1} = f_3^{\boldsymbol{\theta}_3}(I_1) \qquad (1) \\
\tilde{D}_2 &= \frac{(\alpha_2 - \nu_2) I_2 + \delta_2 D_2}{\alpha_2} = f_4^{\boldsymbol{\theta}_4}(I_2)
\end{aligned}
$$

where $\boldsymbol{\theta}_1 = (\beta_{11}, \beta_{12}, \phi_1, \gamma_1)$, $\boldsymbol{\theta}_2 = (\beta_{22}, \beta_{21}, \phi_2, \gamma_2)$, $\boldsymbol{\theta}_3 = (\alpha_1, \nu_1, \delta_1)$, and $\boldsymbol{\theta}_4 = (\alpha_2, \nu_2, \delta_2)$.

Distribution of $\tilde{D}_1, \tilde{D}_2$

We used a bootstrapped data set $(\tilde{d}_i^1,\ \tilde{d}_i^2,\ i = 1, \ldots, B)$ as observed data. Assume that $\tilde{d}_i^1$s and $\tilde{d}_i^2$s are normally distributed, i.e

$$L(D_1, D_2) \propto \frac{1}{\sigma_1 \sigma_2} \exp\left[\frac{1}{2\sigma_1^2} \sum_{i=1}^{B} \left\{\tilde{d}_i^1 - \frac{f_1^{\boldsymbol{\theta}_1}(l_1, l_2) + f_3^{\boldsymbol{\theta}_3}(l_1)}{2}\right\}^2 + \frac{1}{2\sigma_2^2} \sum_{i=1}^{B} \left\{\tilde{d}_i^2 - \frac{f_2^{\boldsymbol{\theta}_2}(l_1, l_2) + f_4^{\boldsymbol{\theta}_4}(l_2)}{2}\right\}^2\right] \quad (2)$$

We assign σ_1^2 and σ_2^2 to be 2.2289, the value of the sample standard deviation of the data.

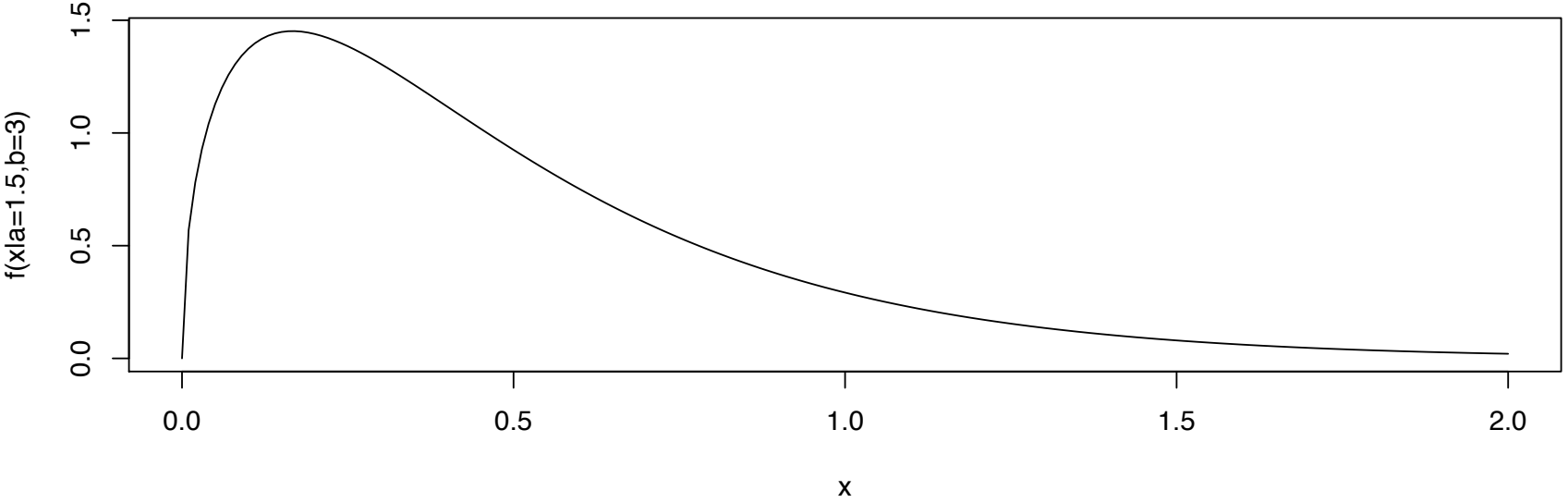
Priors for $\boldsymbol{\theta}_1$, $\boldsymbol{\theta}_2$, $\boldsymbol{\theta}_3$, $\boldsymbol{\theta}_4$

All parameters in parameter sets $\boldsymbol{\theta}_1$, $\boldsymbol{\theta}_2$, $\boldsymbol{\theta}_3$, $\boldsymbol{\theta}_4$ are rate parameters in the ODEs systems and should have positive values. We assign the same gamma prior distribution for all 14 rate parameters. The priors for all rate parameters are given by

$$\pi(\cdot) \sim gamma(a, b) \text{ for all } \boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3, \boldsymbol{\theta}_4. \tag{3}$$

Priors for $\boldsymbol{\theta}_1$, $\boldsymbol{\theta}_2$, $\boldsymbol{\theta}_3$, $\boldsymbol{\theta}_4$

For the hyper prior parameters, we assign shape parameter of 1.5 and scale parameter of 1/3. According to these hyper prior parameters sets, all parameters have positive support and prevent the rate parameters converging to zero or increasing too large (Calderhead and Girolami, 2011).



Prior for l_1, l_2

We need impute l_1 and l_2 during the MCMC simulation so we treat l_1 and l_2 as unknown parameters. l_1 and l_2 are bounded: $0 \leq l_1 \leq \max(\tilde{d}_i^1)$ and $0 \leq l_2 \leq \max(\tilde{d}_i^2)$ respectively.

We put a gamma prior on l_1 and l_2 with hyper parameters for which the 95% confidence interval covers this range.

$$\pi(l_1) \sim gamma(a_{l_1}, b_{l_1}) \tag{4}$$

$$\pi(l_2) \sim gamma(a_{l_2}, b_{l_2}) \tag{5}$$

Conditional posterior for $\boldsymbol{\theta}_1$, $\boldsymbol{\theta}_2$, $\boldsymbol{\theta}_3$, $\boldsymbol{\theta}_4$

$$\pi(\boldsymbol{\theta}_1^*|\boldsymbol{\theta}_2, \boldsymbol{\theta}_3, \boldsymbol{\theta}_4, l_1, l_2, \tilde{d}_i^1)$$

$$\propto \exp\left[\frac{1}{2\sigma_1^2}\sum_{i=1}^{B}\left\{\tilde{d}_i^1 - \frac{\gamma_1^* l_1}{\beta_{11}^* l_1 + \beta_{12}^* l_2 + \phi_1^*}\right.\right. \tag{6}$$

$$\left.\left. -l_1 - \frac{(\alpha_1 - \nu_1)l_1 + \delta_1 D_1}{\alpha_1}\right\}^2\right]$$

$$\times(\beta_{11}^*\beta_{12}^*\phi_1^*\gamma_1^*)^{a-1}\exp\{(\beta_{11}^* + \beta_{12}^* + \phi_1^* + \gamma_1^*)b\}, \tag{7}$$

$$\pi(\boldsymbol{\theta}_2^*|\boldsymbol{\theta}_1, \boldsymbol{\theta}_3, \boldsymbol{\theta}_4, l_1, l_2, \tilde{d}_i^2)$$

$$\propto \exp\left[\frac{1}{2\sigma_2^2}\sum_{i=1}^{B}\left\{\tilde{d}_i^2 - \frac{\gamma_2^* l_2}{\beta_{21}^* l_1 + \beta_{22}^* l_2 + \phi_2^*}\right.\right. \tag{8}$$

$$\left.\left. -l_2 - \frac{(\alpha_2 - \nu_2)l_2 + \delta_2 D_2}{\alpha_2}\right\}^2\right]$$

$$\times(\beta_{21}^*\beta_{22}^*\phi_2^*\gamma_2^*)^{a-1}\exp\{(\beta_{21}^* + \beta_{22}^* + \phi_2^* + \gamma_2^*)b\}, \tag{9}$$

Conditional posterior for l_1, l_2

$$\pi(l_1^*|\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3, \boldsymbol{\theta}_4, l_2, \tilde{d}_i^1)$$

$$\propto \exp\left[\frac{1}{2\sigma_1^2}\sum_{i=1}^{B}\left\{\tilde{d}_i^1 - \frac{\gamma_1 l_1^*}{\beta_{11} l_1^* + \beta_{12} l_2 + \phi_1}\right.\right. \tag{10}$$

$$\left.\left. -l_1^* - \frac{(\alpha_1 - \nu_1) l_1^* + \delta_1 D_1}{\alpha_1}\right\}^2\right]$$

$$\times (l_1^*)^{a_{l_1}-1} \exp\{l_1^* b_{l_1}\}, \tag{11}$$

$$\pi(l_2^*|\boldsymbol{\theta}_1, \boldsymbol{\theta}_2, \boldsymbol{\theta}_3, \boldsymbol{\theta}_4, l_2, \tilde{d}_i^2)$$

$$\propto \exp\left[\frac{1}{2\sigma_2^2}\sum_{i=1}^{B}\left\{\tilde{d}_i^2 - \frac{\gamma_2 l_2^*}{\beta_{21} l_1 + \beta_{22} l_2^* + \phi_2}\right.\right. \tag{12}$$

$$\left.\left. -l_2^* - \frac{(\alpha_2 - \nu_2) l_2^* + \delta_2 D_2}{\alpha_2}\right\}^2\right]$$

$$\times (l_2^*)^{a_{l_2}-1} \exp\{l_2^* b_{l_2} b\}, \tag{13}$$

Proposal for $\boldsymbol{\theta}_1$, $\boldsymbol{\theta}_2$, $\boldsymbol{\theta}_3$, $\boldsymbol{\theta}_4$

The conditional posterior distributions of $\boldsymbol{\theta}_1$, $\boldsymbol{\theta}_2$, $\boldsymbol{\theta}_3$, $\boldsymbol{\theta}_4$ are products of normal likelihood and gamma priors and don't have closed form. Hence, draw samples using Metropolis-Hastings algorithm. We consider multivariate normal distribution as a proposal distribution for conditional posteriors such that

$$\boldsymbol{\theta}_k^* \sim MVN(\boldsymbol{\theta}_k^m, t_k \times \Sigma_k), \quad k = 1, 2, 3, 4,$$

where $\boldsymbol{\theta}^m$ is the current samples; Σ_k is a diagonal matrix; t_k, $k = 1, \ldots, 4$ are tuned to achieve acceptance ratio of between 20% and 40%.

Proposal for l_1, l_2

Similarly, we use independent univariate normal distributions as a proposal distributions for the conditional posteriors of l_1, l_2. The proposal distribution has a mean from the current sample and standard deviation of $\tau_{l_1} \cdot 1$ and $\tau_{l_2} \cdot 1$ for l_1 and l_2 respectively. The τ_{l_1} and τ_{l_2} are tuning constants and are tuned so that the acceptance ratio is about 44%.

MCMC procedure

Now, the MCMC procedure to sample the rate parameters and unobserved value of I_1 and I_2 is given by:

1. Assign suitable initial values of each rate parameters of $\boldsymbol{\theta}_1 = (\beta_{11}, \beta_{12}, \phi_1, \gamma_1)$, $\boldsymbol{\theta}_2 = (\beta_{22}, \beta_{21}, \phi_2, \gamma_2)$, $\boldsymbol{\theta}_3 = (\alpha_1, \nu_1, \delta_1)$, $\boldsymbol{\theta}_4 = (\alpha_2, \nu_2, \delta_2)$, I_1, and I_2.
2. Draw samples for $\boldsymbol{\theta}_1$, $\boldsymbol{\theta}_2$, $\boldsymbol{\theta}_2$, and $\boldsymbol{\theta}_4$ independently using Metropolis-Hastings algorithm.
3. Draw a sample for I_1 and I_2 independently using their conditional posteriors and univariate proposal distribution respectively according to the Metropolis-Hastings algorithm.
4. iterate step 2 and 3 until conergence.

We iterate above MCMC procedure 40,000 times and keep every 40 iteration. We remove first 5000 iteration as burning set. We summarize the posterior statistics.

Results - No Environment

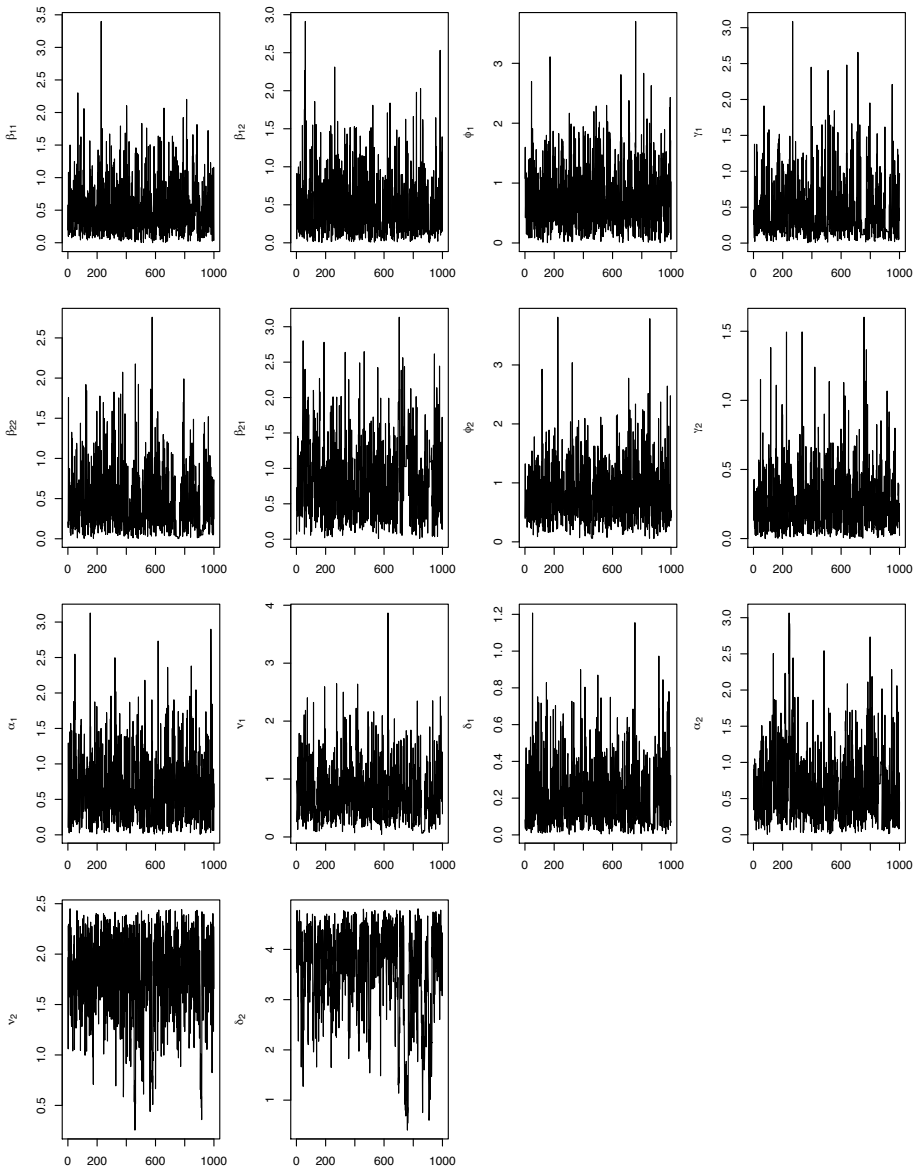


Figure 3

Results - No Environment

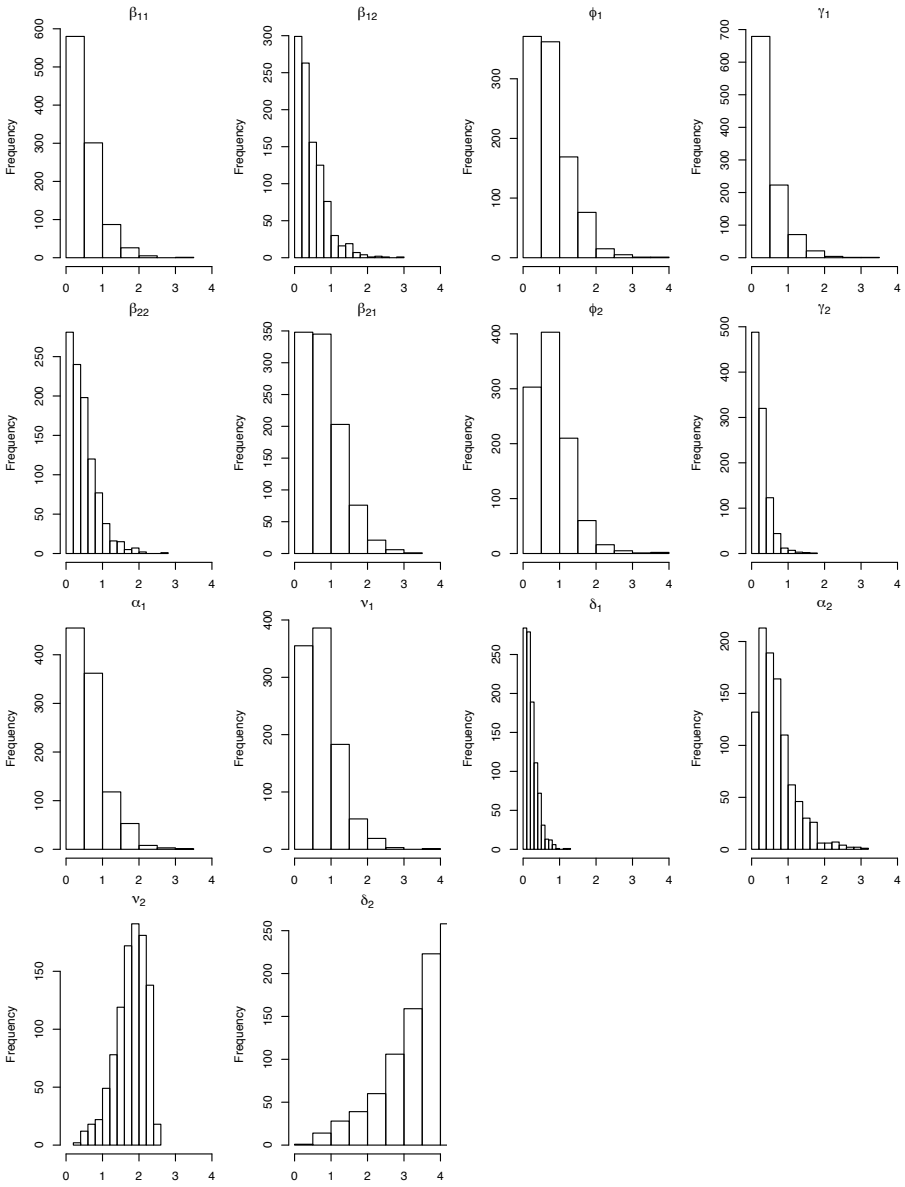


Figure 4

Results - No Environment

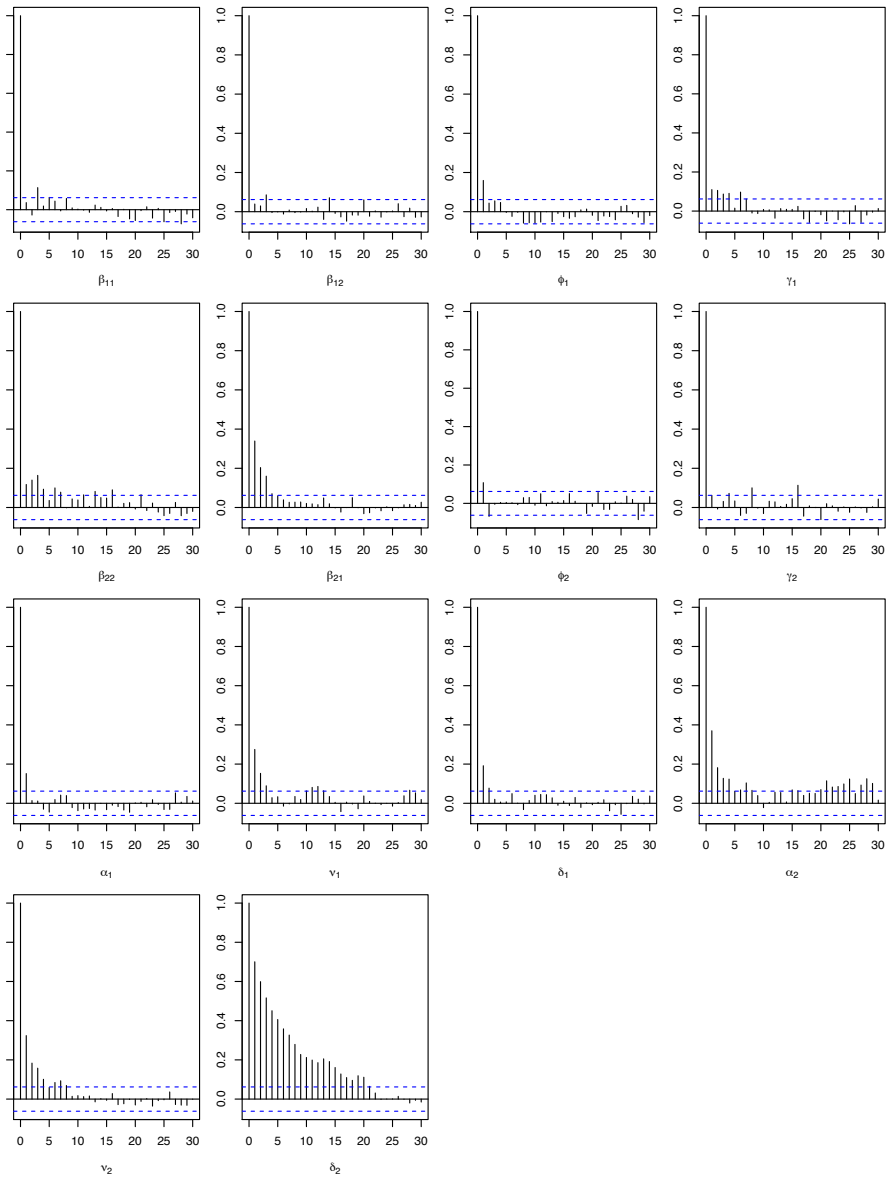


Figure 5

Results - No Environment

	Posterior mean	Posterior SD	L95	U95
β_{11}	0.5316	0.4191	0.0454	1.6545
β_{12}	0.4824	0.4220	0.0313	1.5507
γ_1	0.7024	0.4806	0.0733	1.9213
β_{22}	0.4742	0.4037	0.0429	1.6455
β_{21}	0.4882	0.3670	0.0323	1.3771
γ_2	0.7316	0.5022	0.0824	1.9509
α_1	0.8006	0.4851	0.1282	1.9716
ν_1	0.2656	0.1995	0.0165	0.7505
δ_2	0.6647	0.4867	0.0573	1.9252
α_2	0.7496	0.4683	0.1345	1.9672
ν_2	0.2228	0.1773	0.0199	0.6745
δ_2	0.6649	0.4925	0.0585	1.8735
l_1	1.7448	0.4658	0.6885	2.5085
l_2	3.7009	0.8404	1.6611	4.9509

Table 3: MCMC summary for $n = 50$ and $V = 0$

Results - Environment

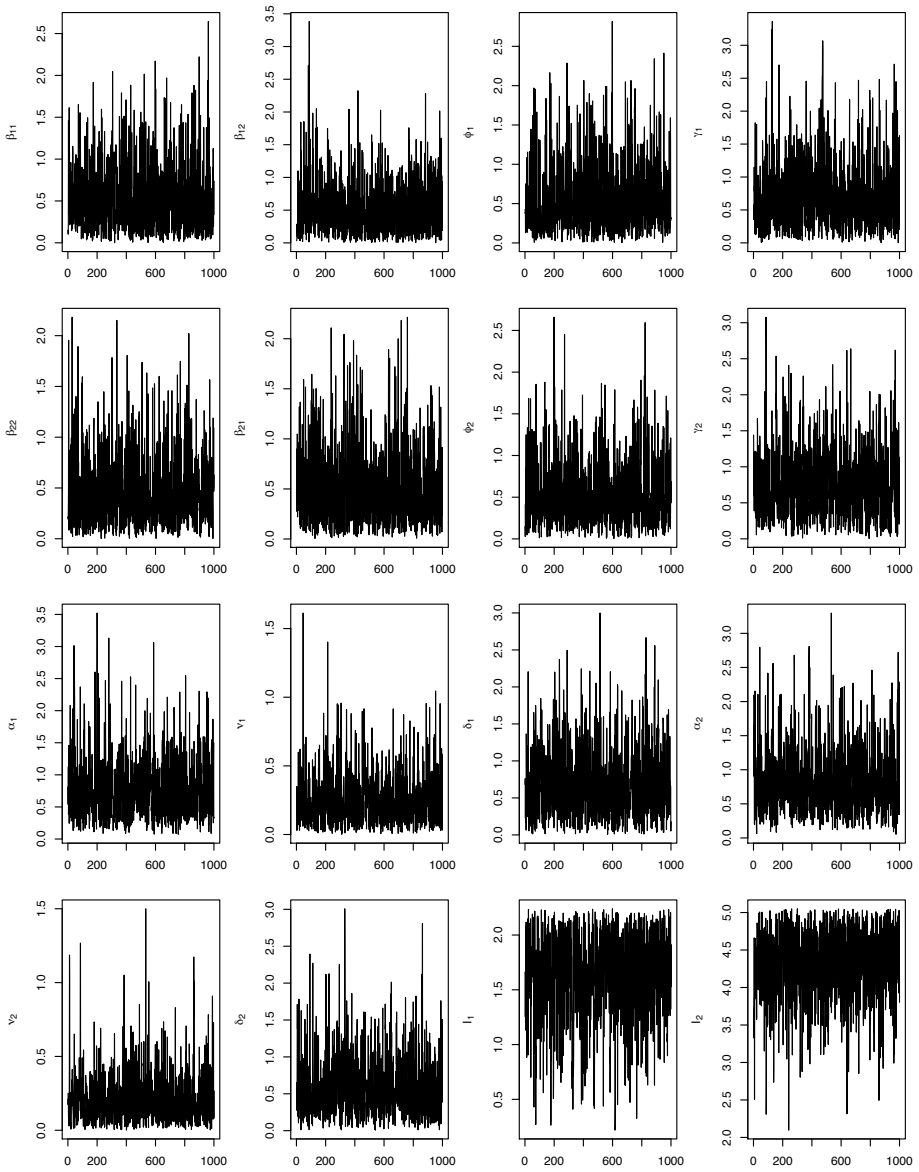


Figure 6

Results - Environment

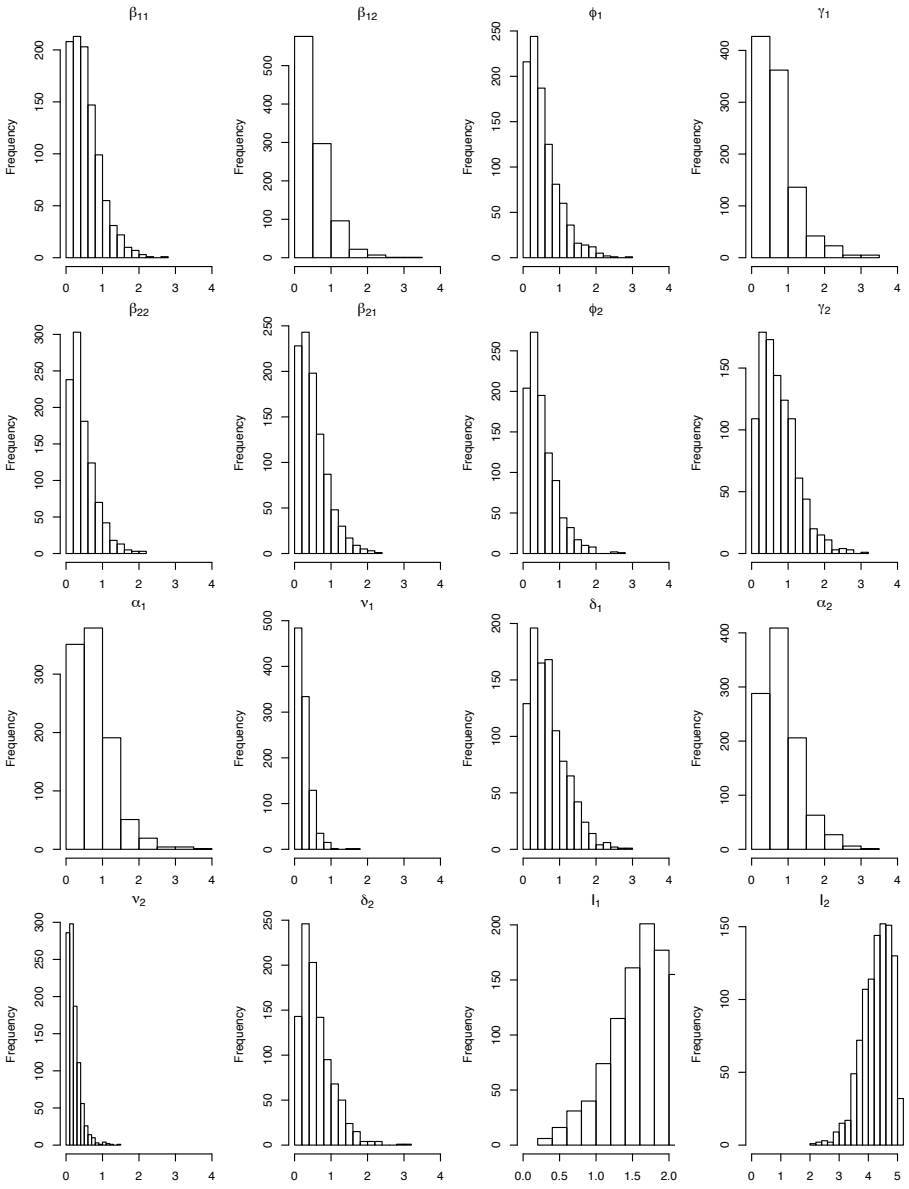


Figure 7

Results - Environment

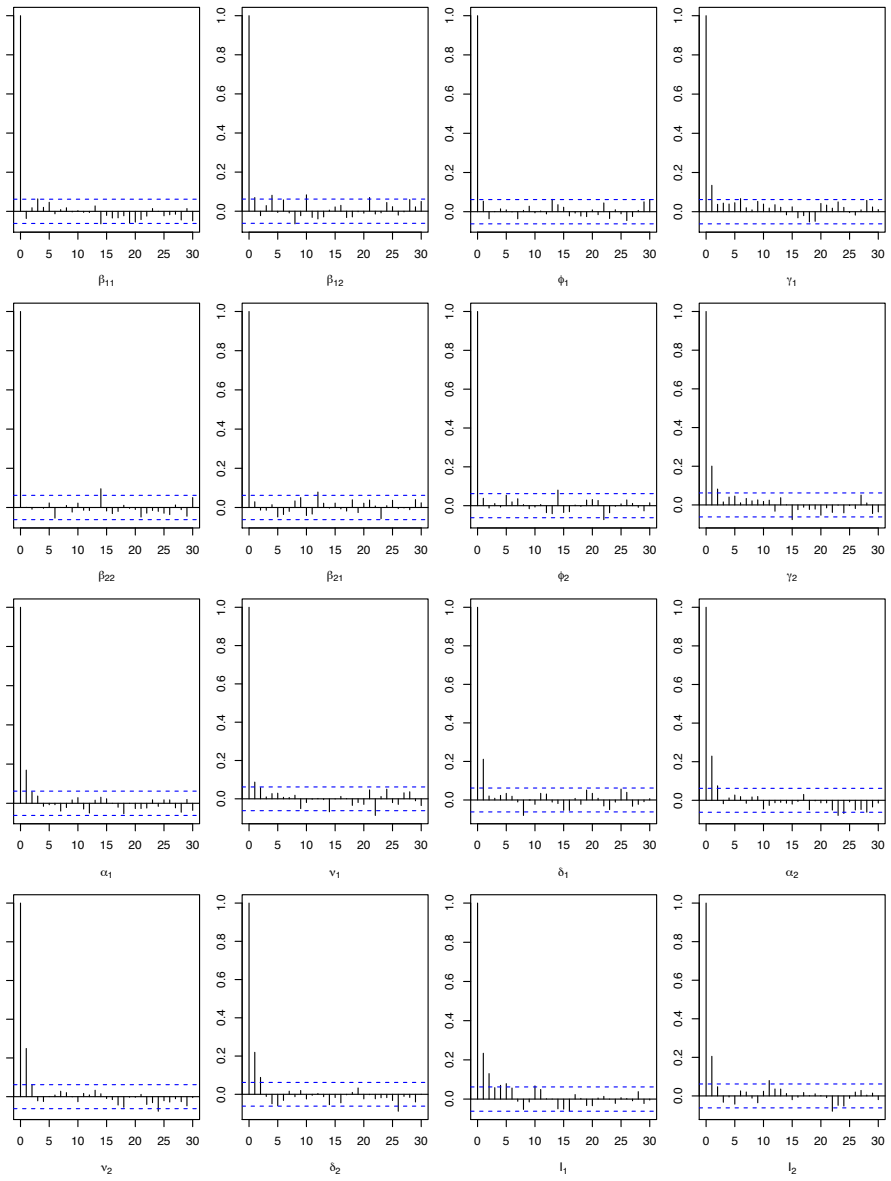


Figure 8

Results - Environment

	Posterior mean	Posterior SD	L95	U95
β_{11}	0.5350	0.4056	0.0366	1.5541
β_{12}	0.5128	0.4041	0.0417	1.5410
ϕ_1	0.5254	0.3991	0.0540	1.5526
γ_1	0.6371	0.4382	0.0798	1.6457
β_{22}	0.4801	0.3796	0.0364	1.5018
β_{21}	0.5266	0.4094	0.0388	1.5610
ϕ_2	0.5537	0.4279	0.0376	1.7793
γ_2	0.7937	0.6352	0.0761	2.3740
α_1	0.7401	0.4461	0.1541	1.8767
ν_1	0.2514	0.1844	0.0333	0.7074
δ_2	0.6994	0.4830	0.0621	1.9595
α_2	0.8236	0.4810	0.1446	2.0287
ν_2	0.2233	0.1777	0.0138	0.6557
δ_2	0.6217	0.4502	0.0561	1.8455
l_1	1.6693	0.3838	0.8531	2.3165
l_2	4.2566	0.6144	2.8644	5.1508

Table 4: MCMC summary for $n = 50$ and $V = 1$

Summary

Using cross-sectional data, we fit a dynamical model of infection. More specifically:

- Determined household structure for a disease from 64 households.
- Used this likelihood and MLE estimates for parameters to generate data for households.
- Derived functional equations for imputing susceptible and infective compartments.
- Used generated averages of households to fit a dynamical model in steady state using a MCMC procedure. This provides point estimates as well as 95% credible intervals for the parameters and imputed states.

Thanks